

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

Statement of Contribution

1. General Role in the Project

I led the technical development of the entire NYC-based data pipeline and played a major role in shaping the predictive and analytical foundation of our project. My work formed the technical backbone of the project, and I was also responsible for writing most of the core sections of the IEEE paper, including the full Related Work, NYC Analysis, Feature Engineering, Predictive Modeling, and Validation Techniques.

2. Technical Contribution

- Developed the initial project codebase, including all shared data processing and analysis functions.
- Defined and structured all NYC datasets, including loading, cleaning, transformation, and merging across multiple sources.
- Applied H3 hexagonal spatial indexing (resolution 9) for neighborhood-level aggregation of NYC air quality data.
- Engineered features (PM2.5, PM10, temperature, humidity, etc.) at the H3-cell level, using z-score normalization.
- Designed and implemented K-Means clustering to identify high/medium/low pollution clusters in NYC.
- Built and tuned multivariate regression models to quantify the impact of weather (temperature, humidity) on PM2.5 levels.
- Conducted bootstrapping and RMSE analysis to validate model reliability and stability.
- Produced multiple plots (e.g., clustering scatter plots, coefficient bar charts, RMSE curves) used throughout the final report and presentation.
- Handled the NYC-specific visualizations and statistical summaries included in the final discussion and results.

3. IEEE Paper Contribution

I authored the following key sections of the IEEE paper:

- **Related Work:** I wrote the entire section, synthesizing 10+ references and articulating the gap our study fills.

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Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

- **NYC Analysis:** Covered all stages from data loading to visualization and modeling of pollution patterns in NYC.
- **Methodology (Aggregation & Feature Engineering):** Documented how data was spatially normalized and features computed.
- **Predictive Analysis:** Described both clustering and regression logic in clear academic terms.
- **Reliability & Validation Techniques:** Wrote about bootstrapping, RMSE, and their role in validating our models.

“Please refer to the blue-highlighted sections in the attached IEEE paper in reference section”

4. GitHub Pull Requests (PRs) & Repo Structure & Organization

- Initiated the shared Jupyter Notebook NYC_Boston_Chicago_Project.ipynb and developed the **complete NYC workflow**.
- Modularized functions for cleaning, H3 aggregation, clustering, and regression to be reused by other teammates.
- Exported and documented all **NYC summary statistics, figures, and CSV outputs**.
- Assisted other team members by validating their outputs against the NYC benchmark and ensuring consistency.
- Uploaded Related Work part.
- Uploaded NYC Analysis, Feature Engineering, Predictive Modeling, and Validation Techniques.

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 **WafaAlyammahi** Add files via upload 1b7f6d1 · now  128 Commits

 Data	update	2 days ago
 Documentation	Add files via upload	now
 NYC_Boston_Chicago_Project.ipynb	update	12 hours ago
 README.md	Update README.md	2 days ago
 nyc_boston_chicago_project.ipynb	Created using Colab	13 hours ago

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 History

Name	Last commit message	Last commit date
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 Formulas Used.docx	Add files via upload	last week
 Full Related Work part.docx	Add files via upload	now
 Previous Research.pdf	Add files via upload	last week
 Related Work.docx	update	2 days ago
 Report.docx	update	2 days ago
 The Selected Topic.docx	update	2 days ago

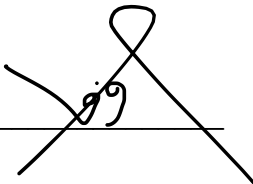
6. Peer Review & Team Involvement

- Helped guide team members through the H3 encoding process and statistical analysis design.
- Participated actively in weekly meetings, coordinated shared responsibilities, and reviewed all merged work.
- Supported documentation and quality control of the overall report and GitHub repository.

Declaration

I confirm this document truthfully represents my contribution to the group project.

Signature:



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Reference:

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

A Comparative Spatial Analysis of Urban Air Quality: New York City, Boston, and Chicago

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ments.

We account for the variations in environmental data through the use of robust validation techniques like bootstrapping and Root Mean Square Error (RMSE) testing to ensure maximum statistical reliability and accuracy of predictions. These techniques strengthen the robustness of findings and ensure greater confidence in the relationships modeled between weather conditions and pollutant concentration.

With intuitive visualizations like heatmaps, scatterplots, and 3D regression surfaces, we turn complex environmental data into actionable insights that inform policymakers, researchers, and public health officials. Our strategy is to answer some of the most fundamental questions regarding urban air quality:

- 1) What are the differences in air pollution concentration patterns in major cities?
- 2) Which cities have the most PM_{2.5} air pollution burdens?
- 3) Which are the strongest indicators of weather conditions that are related to air quality deterioration?
- 4) Can commutative capacity of air quality measures among urban environments be improved by analytical methods based on GIS?

The framework that has been created in this study has data-driven and scalable procedures to carry out integrated air quality assessment in urban settings. Increased accuracy in cross-city comparison and identification of determinants of pollutant attributes is made possible with the proposed approach, allowing evidence-based decision support to frame environmental policy, resource planning, and intervention measures towards improving urban air quality and disease prevention.

II. RELATED WORK

The combination of mobility patterns and environmental conditions, in this case air quality, is an important aspect of grasping how city dynamics create and contribute to climatic changes. With increasing urban expansion, understanding how mobility patterns are affecting the environmental parameters like air pollution becomes relevant. This project extends this line of reasoning by comparing mobility patterns of different cities and determining how these impact air quality. A number of important studies form the basis for this work, providing methodologies and findings that are immediately relevant to what we are doing.

Al Jawarneh et al. [1] introduced a scalable method to combine heterogeneous mobility and air pollution data with QoS guarantees. They have a highly effective model that is capable of efficiently handling big geospatial data to support joint analysis on multi-faceted urban datasets. This directly influences how we're handling mobility and environmental data in such a manner that guarantees the robustness of our analysis, especially while comparing air quality between cities.

Further, Al Jawarneh et al. [2] were concerned with approximate query processing in smart cities, utilizing spatial data streams to process mobility and environmental conditions in real-time. This highlights the importance of effective processing methods capable of withstanding the constant stream of data produced by urban systems. This is in line with our work

since real-time analysis of mobility and air quality will play a crucial role in getting actionable insights that will inform policy measures.

Another pertinent work by Al Jawarneh et al. [3] is on the use of mobility and environmental information in climate change analytics. This work investigates how the mobility of urban areas may be analyzed together with environmental parameters in order to offer a more holistic view of how cities are affected by climate change. The work of theirs emphasizes the need to integrate several sources of data to investigate the intricate association between urban mobility and environmental conditions—an approach that will be followed in our project.

Al Jawarneh et al. [4] also made a valuable contribution by emphasizing the simplification of sophisticated geospatial data towards efficient analytics. The research conducted by them on polygon simplification of georeferenced big data will be useful in the handling of big datasets like those with mobility and air quality data in cities. Simplifying the spatial data will assist us in enhancing the efficiency and scalability of analysis back in our own work.

In addition, the research by Ghaffaripasand et al. [5] explores how urban mobility affects air pollution in Kampala, a sub-Saharan city. The authors integrated mobility data from GPS and air quality monitoring to model transportation patterns against the concentration of pollutants. The study employs statistical modeling in order to estimate how mobility in the urban environment contributes to pollutant concentration. This research is very relevant to our project because it is insightful regarding how mobility patterns influence air quality, especially in cities with heterogeneous urban dynamics.

Bandeira and colleagues [6] analyzed the influence of land use on mobility patterns and emissions in a Portuguese city of medium size. Traffic data collection, land use categorization, and air pollutant monitoring were used in their work to determine how the variety of land use categories impacts mobility and emissions. This approach is very relevant to our research because knowing how land use drives mobility patterns will allow us to compare metropolitan environments in different cities and compare how these environments contribute to air quality.

Furthermore, Austin et al. [7] applied k-means clustering to group U.S. air monitoring sites based on long-term PM_{2.5} composition. They showed how source types and geography affect air pollution profiles. Their study highlighted the value of clustering multivariate pollution data. This informs city-level clustering methods for mobility and air quality patterns.

Ross et al. [8] suggested a hybrid land use regression and temporal adjustment model for estimating exposure to PM_{2.5} and NO₂ across NYC. Their detailed spatial modeling using health outcome data highlights the importance of high-resolution environmental exposure estimation. This parallels our city-level geospatial aggregation using H3 indices.

Liang and Gong [9] conducted a large-scale study across 626 Chinese cities. They analyzed how different urban forms (compact vs. dispersed) influence long-term PM_{2.5} trends.

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

Their findings on the varying impact of urban form metrics depending on city development stages support our intent to compare multiple cities with different urban structures.

Safarov et al. [10] applied advanced GIS-based interpolation and structural equation modeling in Pavlodar and Kazakhstan to quantify spatial pollution patterns and their relationship with environmental variables. Their use of mobile sensors and geospatial modeling illustrated the effectiveness of spatially-resolved air quality analysis. This is a method we replicate using mobility and pollution data from different cities.

Collectively, these research works offer key methodologies and frameworks to synthesize and analyze geospatial big data in urban mobility and environmental settings. They note the challenges and methods involved in handling large-scale data, maintaining real-time speeds, and extracting actionable intelligence from intricate urban data sets. These works are the foundation of how we compare the patterns of mobility across cities and measure the consequence on air quality, and ultimately, what policy recommendations are put forth to enhance urban sustainability.

III. METHODOLOGY

A. Data Collection

1) *Data Source and Features:* To conduct a comparative spatial analysis of New York City, Boston, and Chicago air quality, we downloaded publicly available sensor-based environmental data for each city. The data include major air pollutants (PM2.5, NO, PM10, CO, and O) and contextual environmental variables (temperature, humidity), geospatial coordinates (latitude, longitude), and time stamps.

New York City data was collected from five files with data on PM2.5, PM10, temperature, humidity, and sensor locations. Boston's dataset included PM2.5 and NO levels along with temperature and humidity. Chicago's data, sampled from five components too included the most general pollutant coverage of PM2.5, CO, calibrated NO, and O readings, along with meteorological and device-related variables such as battery and signal strength.

2) *Data Collection Limitations and Challenges:* This study is related to the lack of overlap between pollutant data in the three cities. PM10 data was available only in New York City, and O and CO were available only in Chicago, precluding the possibility of consistent multi-variable comparison across all the datasets. This lack of overlap limited the generalizability of some of the analyses and required city-specific modeling. Moreover, although all datasets included time-stamped measurements, differences in sampling frequency and sensor calibration may have introduced heterogeneity in data quality and temporal accuracy. A further challenge was heterogeneity in the amount of data across different cities, with some sites having a much greater quantity of sensor readings than others, potentially influencing the results. Sparse metadata on sensor types and deployment criteria reduced transparency in data reliability.

To surmount these limitations, we used city-specific models, implemented H3 hexagonal indexing to normalize spatial

comparisons, and relied on statistical validation (e.g., RMSE, bootstrapping) to enhance result reliability.

B. Data Processing and Engineering

1) *Data Cleansing and Outlier Handling:* We developed a comprehensive preprocessing pipeline for time and weather data in raw datasets. Early issues included various time stamp formats, missing values for key variables (e.g., PM2.5, temperature, humidity), and extreme outliers from sensor malfunctions. Time stamps were centralized so cities have a shared time format. Missing values in the dataset were eliminated or imputed via forward-fill and mean imputation.

To ensure validity of data, measurements outside domain-specific ranges were filtered out. These included PM2.5 measurements outside 0–500 $\mu\text{g}/\text{m}^3$, temperatures outside -30°C to 60°C , and humidity outside 0% to 100%. These preprocessing measures were undertaken to ensure the reliability of downstream spatial aggregation, clustering, and regression modeling, and to ensure equitable comparison of air quality trends in New York City, Boston, and Chicago.

2) *Spatial Aggregation Using H3 Indexing:* One of the key challenges in this study is that cities differ significantly in layout, size, sensor density, and climate, making it difficult to compare air quality data fairly across regions. To address this, we applied H3 hexagonal spatial indexing, a technique developed by Uber that transforms geographic coordinates into uniform, area-based spatial units. By encoding all sensor locations into hexagonal cells of equal size, we ensured that data from New York City, Boston, and Chicago could be aggregated and compared within a consistent spatial framework.

We chose H3 resolution 9, representing zones of about 0.105 km^2 , ideal for neighbourhood environmental analysis. As shown in Table I, H3 provides a range of resolutions, each aligned with a specific spatial use case—from city-level overviews to street-level detail. Using this method, each hex cell aggregates air quality and environmental measurements (e.g., PM2.5, temperature, humidity), allowing us to conduct city-wide comparisons without bias from differing urban structures. Algorithm 1 details converting raw data into structured spatial grids for clustering and regression analysis.

TABLE I: H3 Resolution and Spatial Coverage

Resolution	Average Hexagon Area (km^2)	Typical Use Case
7	~ 5.16	City-wide overviews
8	~ 0.74	District-level analysis
9	~ 0.105	Neighborhood-level studies
10	~ 0.015	Street-level granularity
11	~ 0.0021	Building-level assessments

3) *Aggregation and Feature Engineering:* After assigning each observation to its corresponding hexagonal zone using the H3 index at resolution 9, we performed spatial aggregation to generate structured input features for modelling. For each

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

Algorithm 1 H3_SpatialAggregateGrid-Based Indexing and Aggregation

```

1: // Step 1: H3 Index Encoding
2: for each record  $d_i \in D$  do
3:    $h_i \leftarrow \text{geo\_to\_h3}(d_i.\text{latitude}, d_i.\text{longitude}, r)$ 
4:   assign  $h_i$  to  $d_i$ 
5: end for
6: // Step 2: Group by Hex Cell
7: group  $D$  by  $h_i$  to form hex_cells  $H = \{H_1, H_2, \dots, H_m\}$ 

8: // Step 3: Aggregate Environmental Variables
9: for each hex  $H_j \in H$  do
10:  compute:
11:    mean_PM2.5  $\leftarrow \text{avg}(\text{PM}_{2.5} \text{ readings in } H_j)$ 
12:    mean_Temp  $\leftarrow \text{avg}(\text{Temperature readings in } H_j)$ 
13:    mean_Humidity  $\leftarrow \text{avg}(\text{Humidity readings in } H_j)$ 
14: end for
15: return  $H$ 

```

H3 cell h , we computed the average values of environmental indicators using the formula:

$$\bar{x}_h = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

where x_i represents the value of a variable (e.g., PM2.5, temperature) in cell h , and n is the number of observations in that cell. The following features were computed per hexagon: mean PM2.5, mean temperature, mean humidity, mean NO (where available), mean PM10 (NYC only), mean O and CO (Chicago only), standard deviation of PM2.5, and the count of observations per cell.

To prepare the data for machine learning tasks, continuous variables were standardized using z-score normalization, defined as:

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

where μ and σ are the mean and the standard deviation of each variable. This ensured that no feature dominated due to its scale and that clustering and regression analyses were not biased. The resulting feature set provided a robust spatial representation of urban air quality, enabling the discovery of localized pollution patterns and their relationship to meteorological conditions across the three cities.

IV. PREDICTIVE ANALYSIS

A. K-Means Clustering

To identify the spatial patterns of pollution in Chicago, Boston, and New York City, K-Means clustering was performed on the aggregated data at hexagon-level for main environmental attributes: PM10 (or NO where the data is accessible), PM2.5, temperature, and humidity. StandardScaler() was used to scale the features beforehand to compare them. The values of k were given as 3, which came from the application of the Elbow Method so that within-cluster compactness as well as within-cluster separation were achieved.

The K-Means algorithm starts by initializing centroids randomly, and repeatedly assigns each H3 cell to the nearest centroid based on Euclidean distance. The centroids are recalculated as mean of assigned points, and loop until centroids stabilize, or convergence. After the model was trained (KMeans (n_clusters=3, random_state=42)), all H3 cells were labeled with a cluster label. The centroids were inverse-transformed back to original scale with inverse_transform() for interpretability.

The scatter plot shown in Figure 2 were created to show the clusters on PM2.5 and PM10 axes and how they presented clear differences between low-, medium-, and high-pollution zones. Urban and industrial zones were attributed to high-pollution clusters, while lower pollution clusters were found in residential or less urbanized areas. Clustering provided meaningful environmental segmentation between cities and allowed for more localized analyses of air quality behaviors.

B. Regression Analysis

To be able to get quantification of meteorological variables towards air pollution, a multivariate linear regression model was employed. The model predicted PM2.5 levels based on two significant environmental variables: temperature and humidity. Hexagon-aggregated data was employed in order to avoid spatial heteroscedasticity in New York City, Boston, and Chicago.

The regression was conducted employing Ordinary Least Squares (OLS) of the statsmodels package. The target variable was PM2.5, and temperature and humidity were employed as independent predictors. The variables were standardized prior to fitting to prevent scale dominance and facilitate interpretability. The fit model produced coefficient estimates, p-values, and the R-squared metric to measure statistical significance and predictability of every feature.

The regression equation was of the form:

$$\text{PM2.5} = \beta_0 + \beta_1 \times \text{Temperature} + \beta_2 \times \text{Humidity} + \varepsilon \quad (3)$$

where β_0 is the intercept term, β_1 and β_2 are the temperature and humidity coefficients respectively, and ε is the residual error term.

To aid comprehension of model performance, a 3D visualization was created that showed actual PM2.5 values and the regression surface within the temperature-humidity space. This was to confirm that the model can generalize trends and illustrated how meteorological variations correlate with the level of PM2.5 pollution in urban areas.

V. RELIABILITY AND VALIDATION TECHNIQUES

To ensure the stability of the regression model and validate the findings, two methods of validation were used: bootstrapping and Root Mean Square Error (RMSE) estimation.

Bootstrapping was applied to examine the stability of the regression coefficients. The method involved resampling the dataset with replacement repeatedly (e.g., 1000 iterations) and reestimating the regression model for each new sample. By looking at the distribution of the estimated coefficients

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

```
# STEP 1: Install Compatible h3 and Required Libraries
!pip install h3==3.7.6
!pip install geopandas matplotlib seaborn scikit-learn statsmodels

# STEP 2: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import h3
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from sklearn.preprocessing import StandardScaler
from sklearn.utils import resample
from sklearn.cluster import KMeans
import statsmodels.api as sm
from google.colab import files

Collecting h3==3.7.6
  Using cached h3-3.7.6-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.8 kB)
Using cached h3-3.7.6-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.1 MB)
Installing collected packages: h3
Successfully installed h3-3.7.6
Requirement already satisfied: geopandas in /usr/local/lib/python3.11/dist-packages (1.0.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.11/dist-packages (0.13.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.11/dist-packages (1.6.1)
Requirement already satisfied: statsmodels in /usr/local/lib/python3.11/dist-packages (0.14.4)
Requirement already satisfied: numpy>=1.22 in /usr/local/lib/python3.11/dist-packages (from geopandas) (2.0.2)
Requirement already satisfied: pyogrio>=0.7.2 in /usr/local/lib/python3.11/dist-packages (from geopandas) (0.10.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from geopandas) (24.2)
Requirement already satisfied: pandas>=1.4.0 in /usr/local/lib/python3.11/dist-packages (from geopandas) (2.2.2)
Requirement already satisfied: pyproj>=3.3.0 in /usr/local/lib/python3.11/dist-packages (from geopandas) (3.7.1)
Requirement already satisfied: shapely>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from geopandas) (2.1.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (4.57.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (1.4.8)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (11.2.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.11/dist-packages (from scikit-learn) (1.15.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.11/dist-packages (from scikit-learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.11/dist-packages (from statsmodels) (1.0.1)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.4.0->geopandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.4.0->geopandas) (2025.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-packages (from pyogrio>=0.7.2->geopandas) (2025.1.31)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
```

✓ NYC Datasets

```
# STEP 3: Upload Files
print("Upload: NYC_PM.csv, NYC_PM_Part1.csv, NYC_PM_Part2.csv, NYC_PM_Part3.csv, NYC_AQ.csv")
uploaded = files.upload()

Upload: NYC_PM.csv, NYC_PM_Part1.csv, NYC_PM_Part2.csv, NYC_PM_Part3.csv, NYC_AQ.csv
Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving NYC_AQ.csv to NYC_AQ.csv
Saving NYC_PM_Part1.csv to NYC_PM_Part1.csv
Saving NYC_PM_Part2.csv to NYC_PM_Part2.csv
Saving NYC_PM_Part3.csv to NYC_PM_Part3.csv
Saving NYC_PM.csv to NYC_PM.csv

# STEP 4: Load and Merge PM Data
pm_main = pd.read_csv("NYC_PM.csv")
pm_parts = [pd.read_csv(f) for f in ["NYC_PM_Part1.csv", "NYC_PM_Part2.csv", "NYC_PM_Part3.csv"]]
pm_all = pd.concat([pm_main] + pm_parts, ignore_index=True)

# STEP 5: Load AQ and Convert Time
aq = pd.read_csv("NYC_AQ.csv")
pm_all["datetime"] = pd.to_datetime(pm_all["time"], unit="s")
aq["datetime"] = pd.to_datetime(aq["time"], unit="s")
```

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

```
# STEP 6: Merge Datasets
merged = pd.merge(pm_all, aq, on=["SensorID", "datetime"], suffixes=("_pm", "_aq"))

# STEP 7: H3 Geohash Encoding
def latlon_to_h3(row, resolution=9):
    return h3.geo_to_h3(row["latitude_pm"], row["longitude_pm"], resolution)

merged["h3"] = merged.apply(latlon_to_h3, axis=1)

# STEP 8: Normalize Features
scaler = StandardScaler()
merged[["pm25_pm_norm", "pm10_norm"]] = scaler.fit_transform(merged[["pm25_pm", "pm10"]].fillna(0))

# STEP 9: Aggregate by H3 Cell
agg_by_h3 = merged.groupby("h3").agg({
    "pm25_pm": ["mean", "std"],
    "pm10": ["mean", "std"],
    "temperature_pm": "mean",
    "humidity_pm": "mean",
    "SensorID": "count"
}).reset_index()
agg_by_h3.columns = ['h3', 'pm25_mean', 'pm25_std', 'pm10_mean', 'pm10_std', 'temp_mean', 'humidity_mean', 'samples']

# STEP 10: EDA - Histograms (using PM data only for PM10)

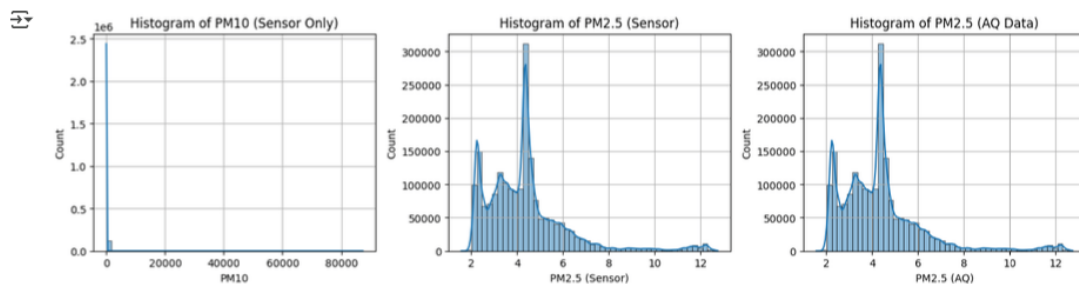
plt.figure(figsize=(14, 4))

# PM10 (from PM sensor only)
plt.subplot(1, 3, 1)
sns.histplot(pm_all["pm10"].dropna(), bins=50, kde=True)
plt.title("Histogram of PM10 (Sensor Only)")
plt.xlabel("PM10")
plt.grid(True)

# PM2.5 from PM sensor
plt.subplot(1, 3, 2)
sns.histplot(merged["pm25_pm"].dropna(), bins=50, kde=True)
plt.title("Histogram of PM2.5 (Sensor)")
plt.xlabel("PM2.5 (Sensor)")
plt.grid(True)

# PM2.5 from AQ source
plt.subplot(1, 3, 3)
sns.histplot(merged["pm25_aq"].dropna(), bins=50, kde=True)
plt.title("Histogram of PM2.5 (AQ Data)")
plt.xlabel("PM2.5 (AQ)")
plt.grid(True)

plt.tight_layout()
plt.show()
```



Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

```
plt.figure(figsize=(6, 4))
filtered_pm10 = pm_all["pm10"].dropna()
```

<https://colab.research.google.com/drive/1mmYImHGw4AM8NeGagd57xfBDaHrPDvxA#scrollTo=6zkGJMYcivsb&printMode=true>

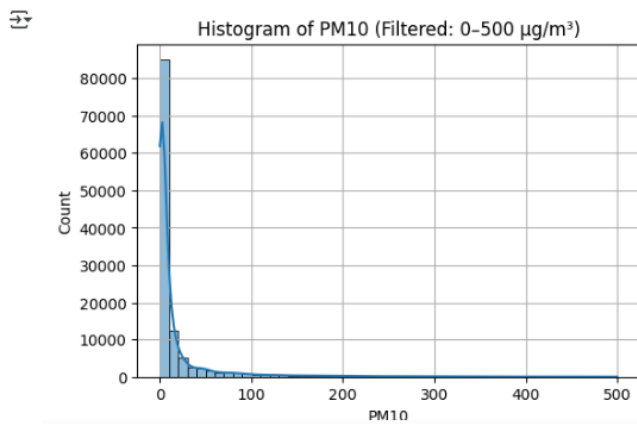
2/17

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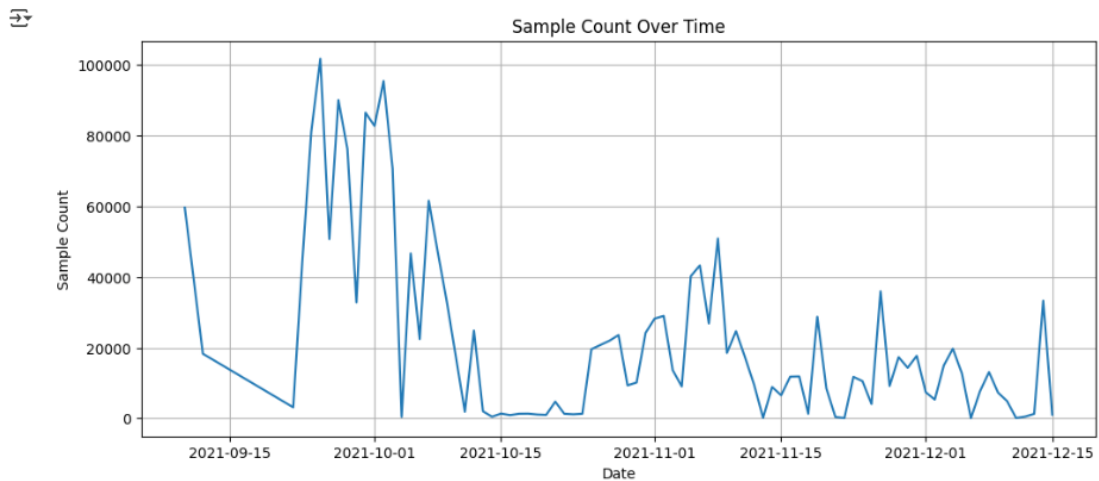
```
filtered_pm10 = filtered_pm10[(filtered_pm10 >= 0) & (filtered_pm10 <= 500)] # Adjust if needed
```

```
sns.histplot(filtered_pm10, bins=50, kde=True)
plt.title("Histogram of PM10 (Filtered: 0-500 µg/m³)")
plt.xlabel("PM10")
plt.grid(True)
plt.show()
```



```
# STEP 11: Time vs Sample Count
sample_over_time = merged.groupby(merged['datetime'].dt.date).size()
```

```
plt.figure(figsize=(12, 5))
sample_over_time.plot()
plt.title("Sample Count Over Time")
plt.xlabel("Date")
plt.ylabel("Sample Count")
plt.grid(True)
plt.show()
```



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```
# STEP 12: Bootstrapping Mean PM10 (from pm_all, filtered)
boot_means = [
    resample(
        pm_all["pm10"][ (pm_all["pm10"] >= 0) & (pm_all["pm10"] <= 500)].dropna(),
        n_samples=1000
    ).mean()
    for _ in range(500)
]

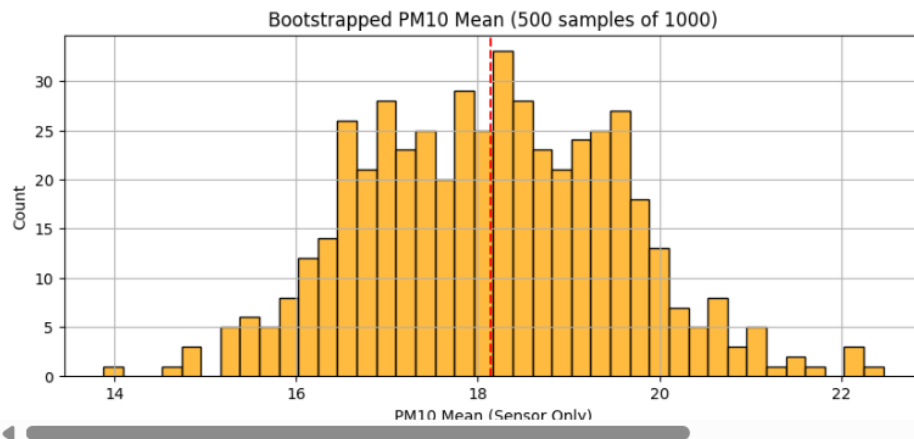
plt.figure(figsize=(10, 4))
sns.histplot(boot_means, bins=40, color='orange')
```

<https://colab.research.google.com/drive/1mmYImHGw4AM8NeGagd57xfBDaHrPDvxA#scrollTo=6zkGJMYcivsb&printMode=true>

4/30/25, 10:59 PM

NYC_Boston_Chicago.ipynb - Colab

```
plt.axvline(np.mean(boot_means), color='red', linestyle='--')
plt.title("Bootstrapped PM10 Mean (500 samples of 1000)")
plt.xlabel("PM10 Mean (Sensor Only)")
plt.grid(True)
plt.show()
```

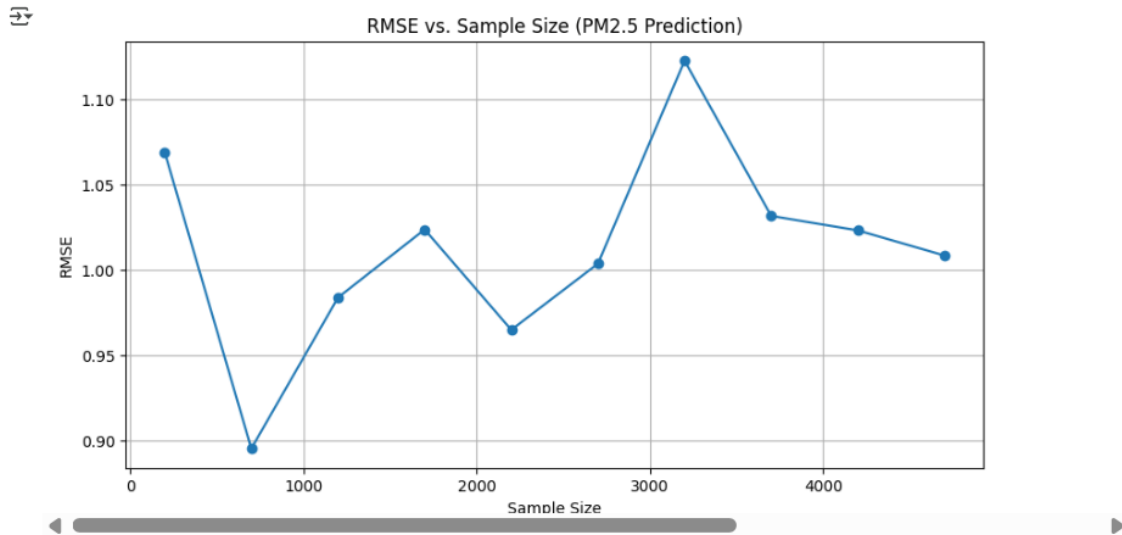


```
# STEP 13: RMSE vs. Sample Size (PM2.5 Prediction)
samples = list(range(200, min(5000, len(merged)), 500))
rmse = []
```

```
for size in samples:
    sample = merged.sample(n=size)
    X = sample[["pm25_pm"]].fillna(0)
    y = sample[["pm25_aq"]].fillna(0)
    model = LinearRegression().fit(X, y)
    preds = model.predict(X)
    rmse = np.sqrt(mean_squared_error(y, preds))
    rmse.append(rmse)

plt.figure(figsize=(10, 5))
plt.plot(samples, rmse, marker='o')
plt.title("RMSE vs. Sample Size (PM2.5 Prediction)")
plt.xlabel("Sample Size")
plt.ylabel("RMSE")
plt.grid(True)
plt.show()
```

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```
# STEP 14: Summary Statistics
summary_stats = merged[["pm25_pm", "pm25_aq"]].describe().T[["mean", "std", "min", "max"]]
summary_stats.loc["pm10_sensor"] = pm_all["pm10"][{(pm_all["pm10"] >= 0) & (pm_all["pm10"] <= 500)}].describe()[["mean", "std", "min"]]
```

<https://colab.research.google.com/drive/1mmYImHGw4AM8NeGagd57xIBDaHrPDvxA#scrollTo=6zkGJMYcivsb&printMode=true>

4/

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```
print("Summary Statistics:")
display(summary_stats)
```

```
Summary Statistics:
```

	mean	std	min	max
pm25_pm	4.325431	1.800967	1.568473	12.744419
pm25_aq	4.325385	1.800867	1.568473	12.744419
pm10_sensor	18.069142	45.071864	0.000000	499.650000

```
# STEP 15: KMeans Clustering (on valid PM sensor data only)

# Recalculate H3 on pm_all to ensure consistency
pm_all["h3"] = pm_all.apply(lambda row: h3.geo_to_h3(row["latitude"], row["longitude"], 9), axis=1)

# Aggregate by H3 using only valid PM10 range
agg_by_h3 = pm_all.groupby("h3").agg({
    "pm25": "mean", # Use "pm25_pm" if needed instead
    "pm10": lambda x: x[(x >= 0) & (x <= 500)].mean(),
    "temperature": "mean",
    "humidity": "mean"
}).reset_index()

agg_by_h3.columns = ["h3", "pm25_mean", "pm10_mean", "temp_mean", "humidity_mean"]

# Filter out NaNs before clustering
features = ["pm25_mean", "pm10_mean", "temp_mean", "humidity_mean"]
X = agg_by_h3[features].dropna().reset_index(drop=True)
X_scaled = StandardScaler().fit_transform(X)

# Apply KMeans clustering
kmeans = KMeans(n_clusters=3, random_state=42)
cluster_labels = kmeans.fit_predict(X_scaled)
```

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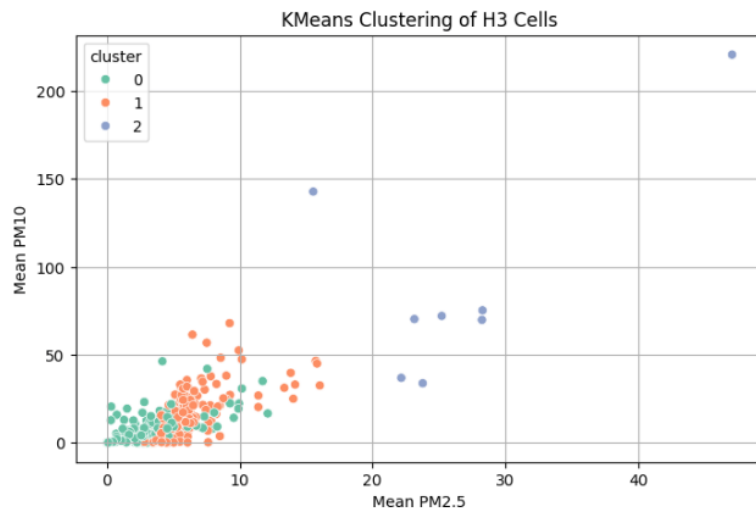
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```
# Create clustered DataFrame
agg_clustered = agg_by_h3.dropna(subset=features).reset_index(drop=True)
agg_clustered["cluster"] = cluster_labels

# Visualize clusters
plt.figure(figsize=(8, 5))
sns.scatterplot(data=agg_clustered, x="pm25_mean", y="pm10_mean", hue="cluster", palette="Set2")
plt.title("KMeans Clustering of H3 Cells")
plt.xlabel("Mean PM2.5")
plt.ylabel("Mean PM10")
plt.grid(True)
plt.show()

# Show cluster centers
cluster_centers = pd.DataFrame(StandardScaler().fit(X).inverse_transform(kmeans.cluster_centers_), columns=features)
cluster_centers["cluster"] = range(3)
print("Cluster Profiles:")
display(cluster_centers)
```

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Cluster Profiles:

	pm25_mean	pm10_mean	temp_mean	humidity_mean	cluster
0	2.480485	5.268230	8.810725	54.818706	0
1	6.252662	16.854878	17.158615	54.079573	1
2	26.686068	90.179471	10.419346	77.511653	2

```
# STEP 16: Multivariate Regression
df = pm_all[["pm10", "pm25", "temperature", "humidity"]].dropna()
df = df[(df["pm10"] >= 0) & (df["pm10"] <= 500)]

X_reg = sm.add_constant(df[["pm25", "temperature", "humidity"]])
y_reg = df["pm10"]

model = sm.OLS(y_reg, X_reg).fit()
print(model.summary())
```



```
OLS Regression Results
=====
Dep. Variable:      pm10      R-squared:                0.322
Model:              OLS      Adj. R-squared:            0.322
Method:             Least Squares      F-statistic:          1.870e+04
Date:               Wed, 30 Apr 2025    Prob (F-statistic):      0.00
Time:               16:16:24      Log-Likelihood:        -5.9520e+05
No. Observations:   118258          AIC:                  1.190e+06
Df Residuals:       118254          BIC:                  1.190e+06
Df Model:           3
Covariance Type:    nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          8.8475         0.288        30.713      0.000         8.283         9.412
pm25           1.8700         0.008       221.304      0.000         1.853         1.887
temperature    -0.0022         0.003        -0.634      0.526        -0.009         0.005
humidity       -0.0731         0.006       -12.960      0.000        -0.084        -0.062
=====
Omnibus:            126401.032    Durbin-Watson:           1.685
Prob(Omnibus):      0.000      Jarque-Bera (JB):        8724015.559
Skew:               5.526      Prob(JB):                 0.00
Kurtosis:           43.600      Cond. No.                 148.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
# STEP 17: Save for Comparison
summary_stats = df.describe()[["pm25", "pm10"]].T[["mean", "std", "min", "max"]]
summary_stats["city"] = "NYC"
agg_clustered["city"] = "NYC"

summary_stats.to_csv("nyc_summary_stats.csv", index=True)
agg_clustered.to_csv("nyc_aggregated_h3.csv", index=False)
```